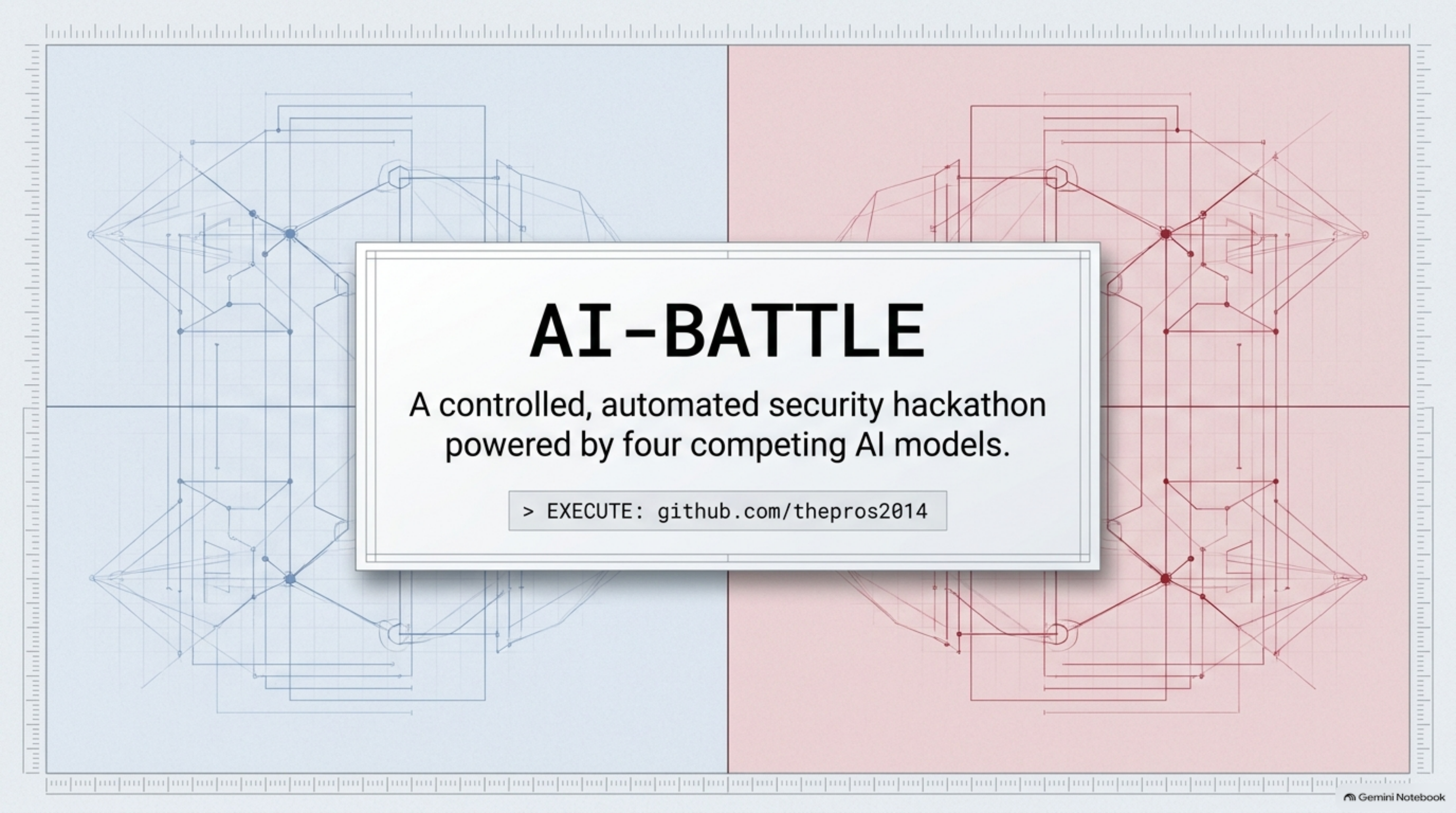


The background is a stylized graphic divided into four quadrants by a light gray cross. The top-left and bottom-left quadrants are light blue, while the top-right and bottom-right quadrants are light red. Each quadrant contains a complex, abstract network of lines and nodes, resembling a circuit board or a neural network diagram. The lines are thin and connect various points, creating a sense of interconnectedness and flow. The overall aesthetic is technical and futuristic.

AI-BATTLE

A controlled, automated security hackathon
powered by four competing AI models.

> EXECUTE: github.com/thepros2014

An ephemeral battleground for autonomous code security.

The Engine

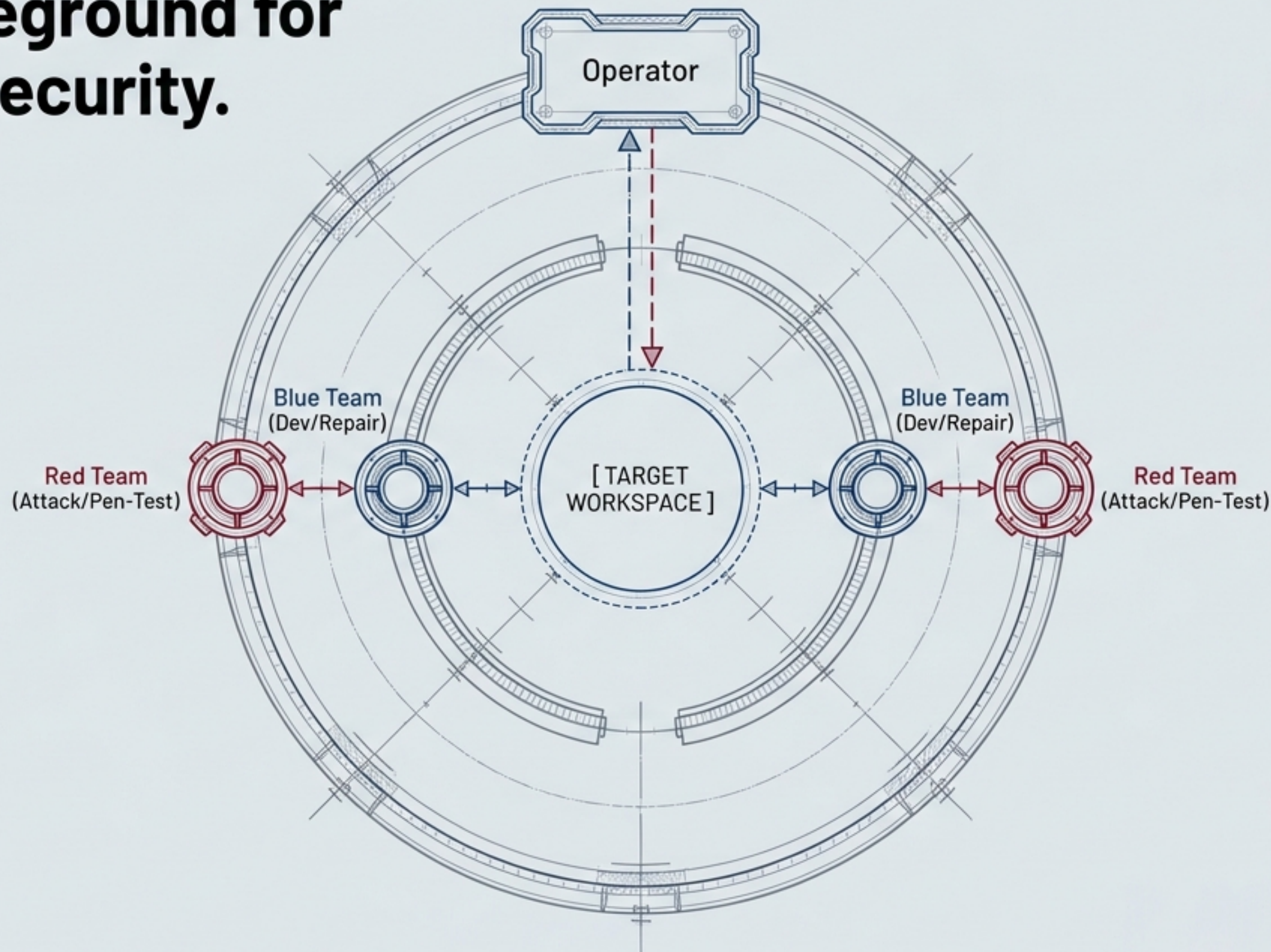
Runs entirely on local hardware, orchestrating the environment without exposing production secrets.

The Combatants

Two paired AI teams autonomously attacking and patching a localized software snapshot.

The Objective

Advance exactly one team at a time through valid, structured adversarial loops.



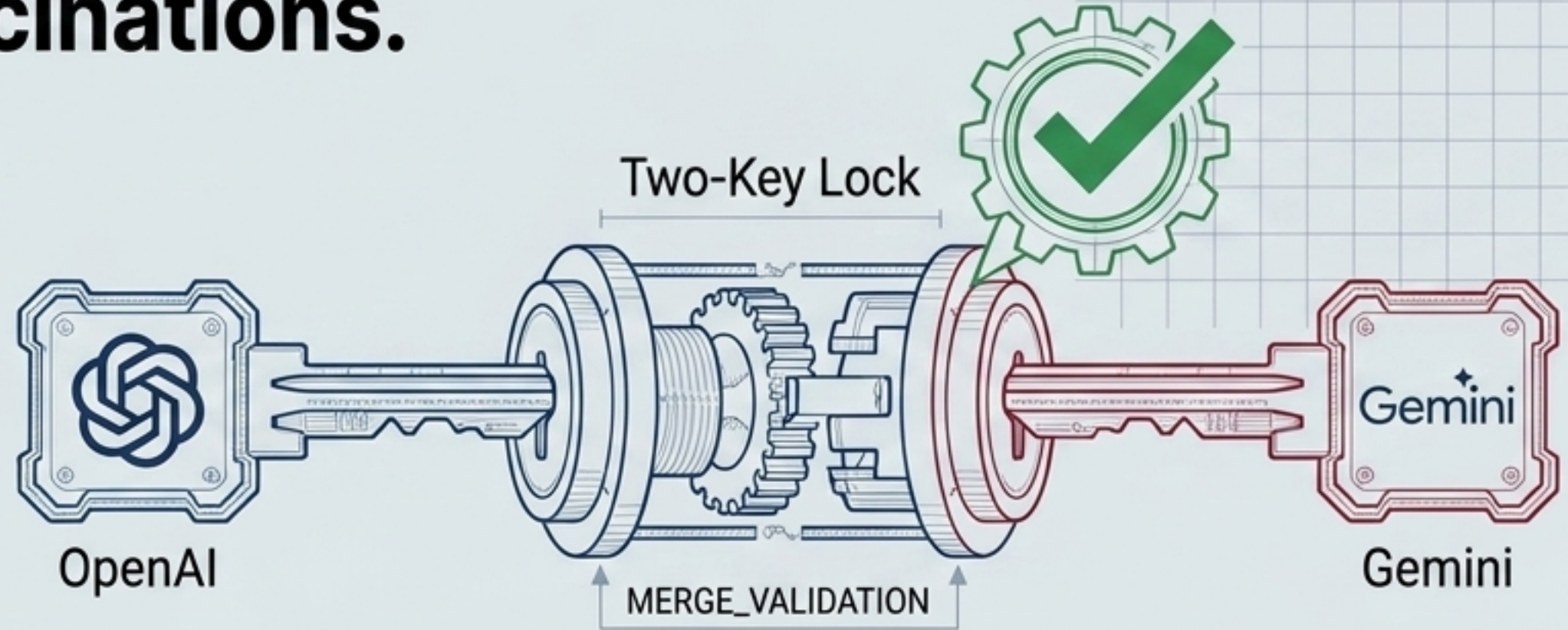
Operational parameters for the Red and Blue teams.

BLUE TEAM	RED TEAM
1 OpenAI Agent + 1 Gemini Agent	1 OpenAI Agent + 1 Gemini Agent
Development, patching, and repair.	Adversarial penetration-testing.
Reads both working/ and breakage/ directories.	Reads ONLY working/ (strictly blind to past findings).
Clears breakage via valid code merge.	Generates valid JSON vulnerability reports.

Multi-model consensus eliminates AI hallucinations.

A single AI agent is prone to hallucinations and phantom code generation.

AI-Battle mitigates this by requiring consensus across rival model architectures.



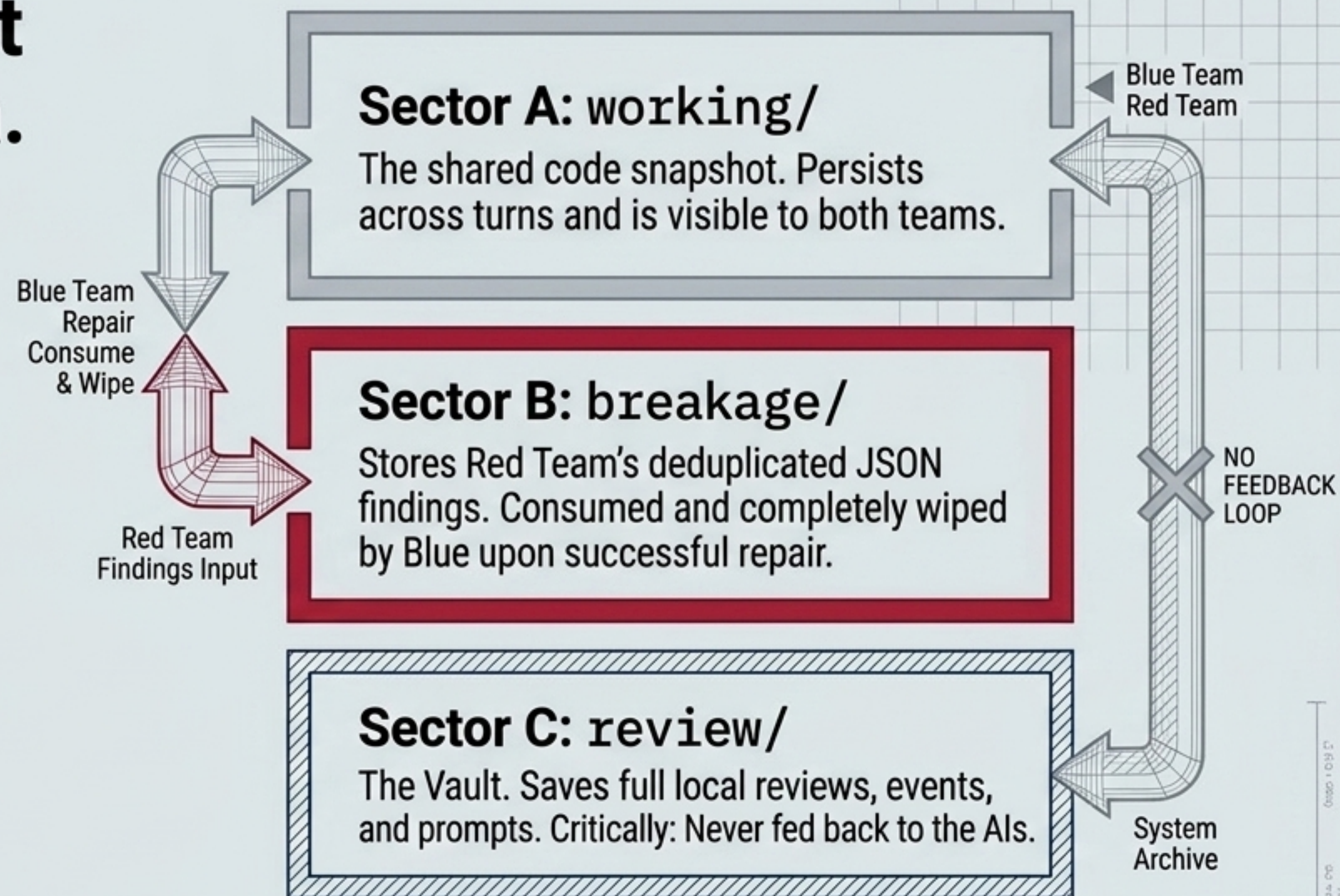
The Rule

Blue team proposals are **ONLY** merged into the codebase if both the OpenAI and Gemini agents agree on every single file they touch.

The Failsafe

Conflicting proposals remain unreleased. The system forces the Blue team to retry the turn until a consensus patch is achieved.

The physical layout of the digital arena.



TACTICAL DIRECTIVE: To avoid context-window bloat, agents navigate sectors using `list_files`, `search_files`, and `chunked read_file` instead of full directory uploads.

The lifecycle of a single tactical engagement loop.

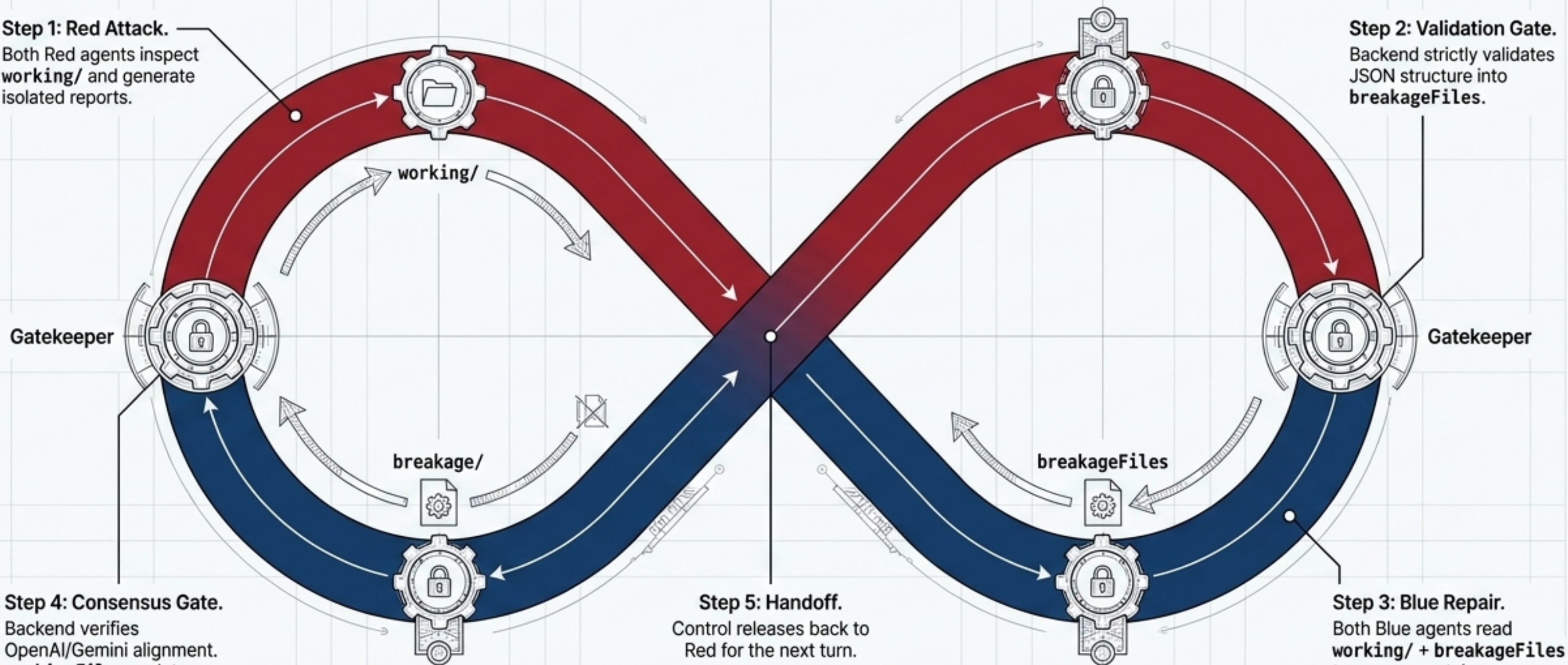
Step 1: Red Attack.
Both Red agents inspect **working/** and generate isolated reports.

Step 2: Validation Gate.
Backend strictly validates JSON structure into **breakageFiles**.

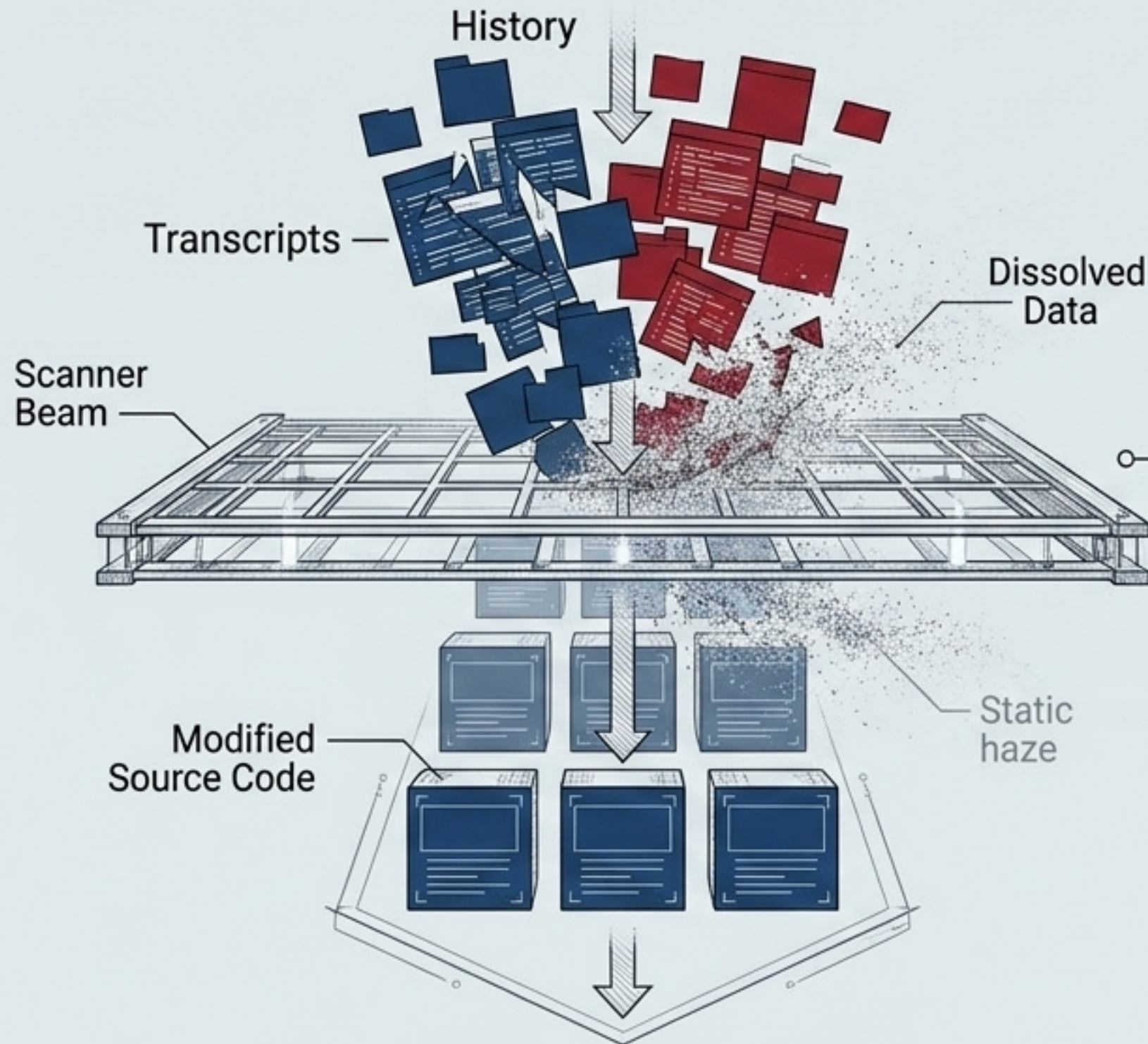
Step 3: Blue Repair.
Both Blue agents read **working/ + breakageFiles** to propose patches.

Step 5: Handoff.
Control releases back to Red for the next turn.

Step 4: Consensus Gate.
Backend verifies OpenAI/Gemini alignment. **workingFiles** updates, **breakage/** is wiped.



Absolute isolation through the Fresh Turn Protocol.



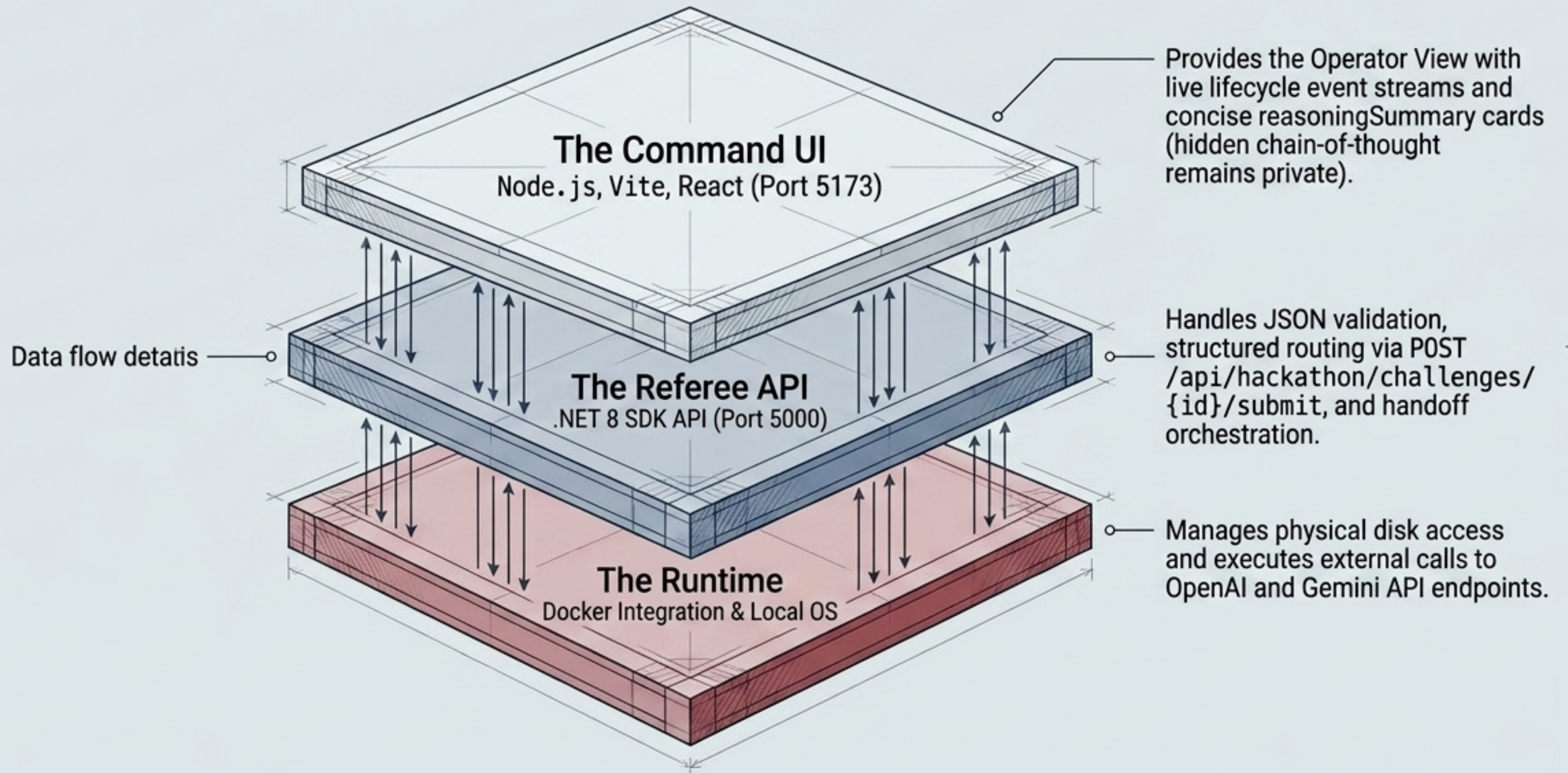
[RULE_01] Provider-side storage is explicitly disabled where supported.

[RULE_02] No previous response IDs or transcripts are utilized in subsequent prompts.

[RULE_03] Active challenge coordination occurs entirely in-memory.

Impact Summary: This forced amnesia prevents context-window bloat and strictly isolates the strategic thinking of the opposing teams. Local artifacts remain safely on disk without polluting future AI context.

A lightweight, locally hosted orchestration engine.



Booting the arena and securely ingesting targets.

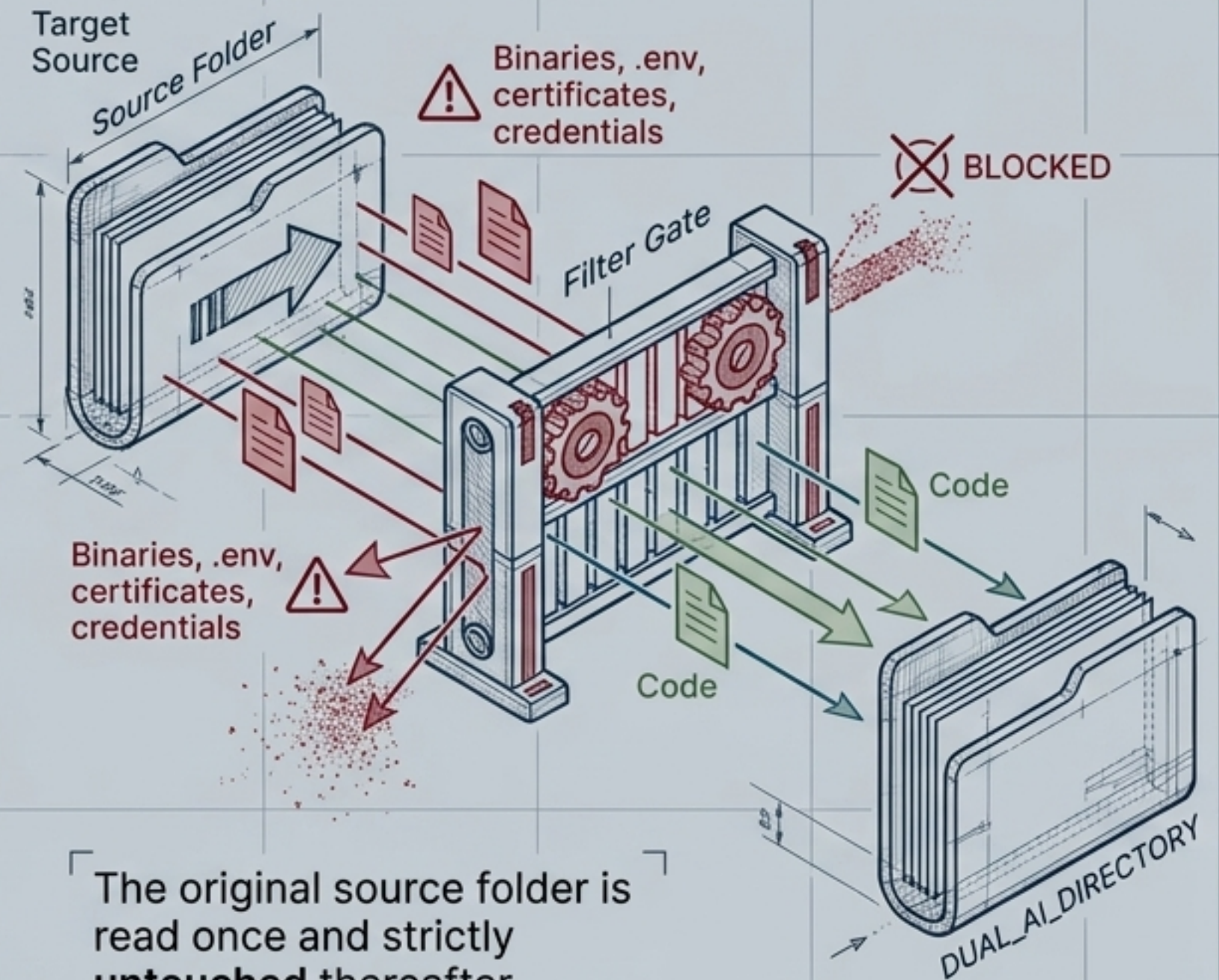
```
> Copy-Item .env.example .env
```

(Holds OPENAI_API_KEY & GEMINI_API_KEY. Never committed to Git.)

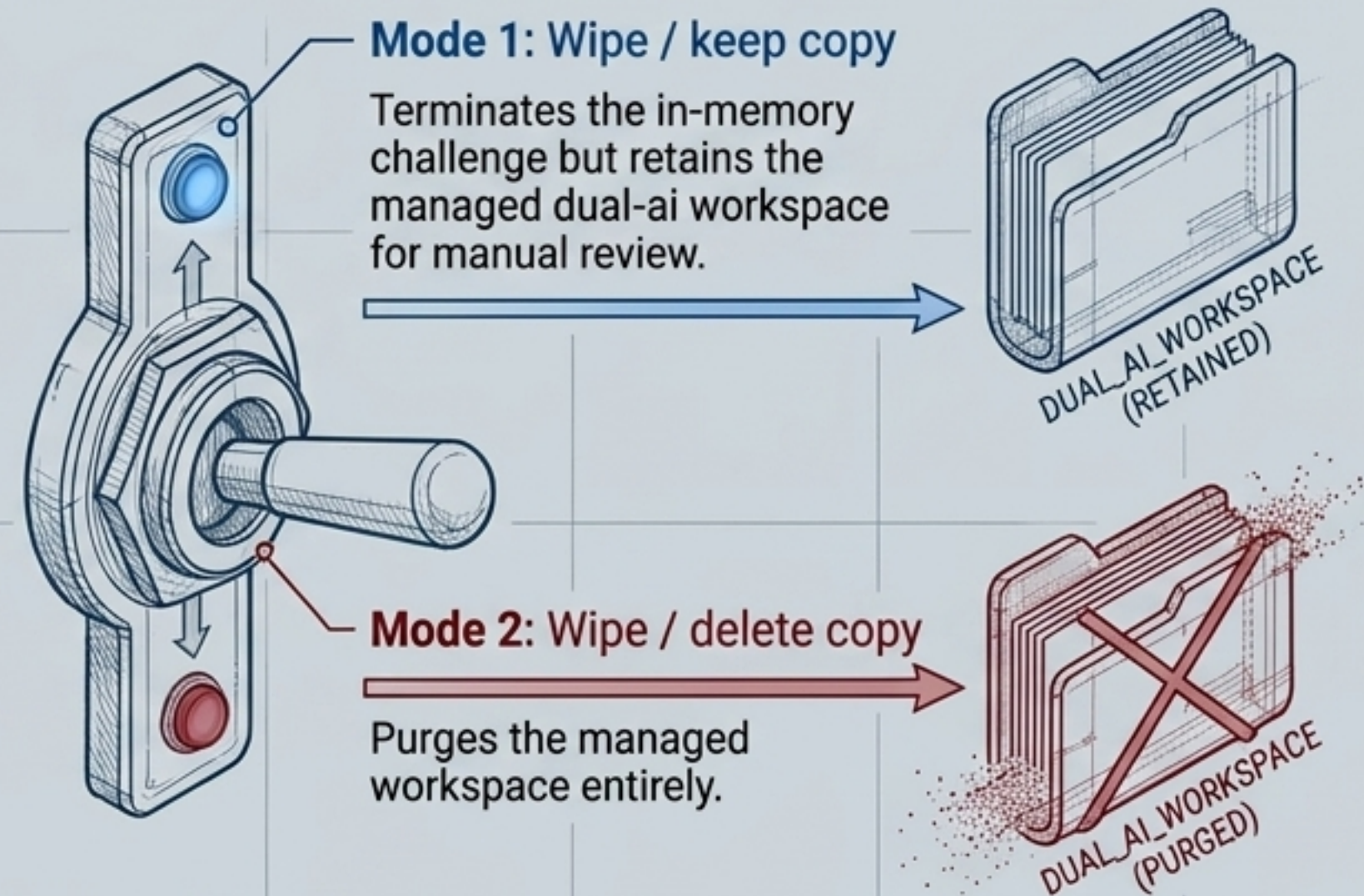
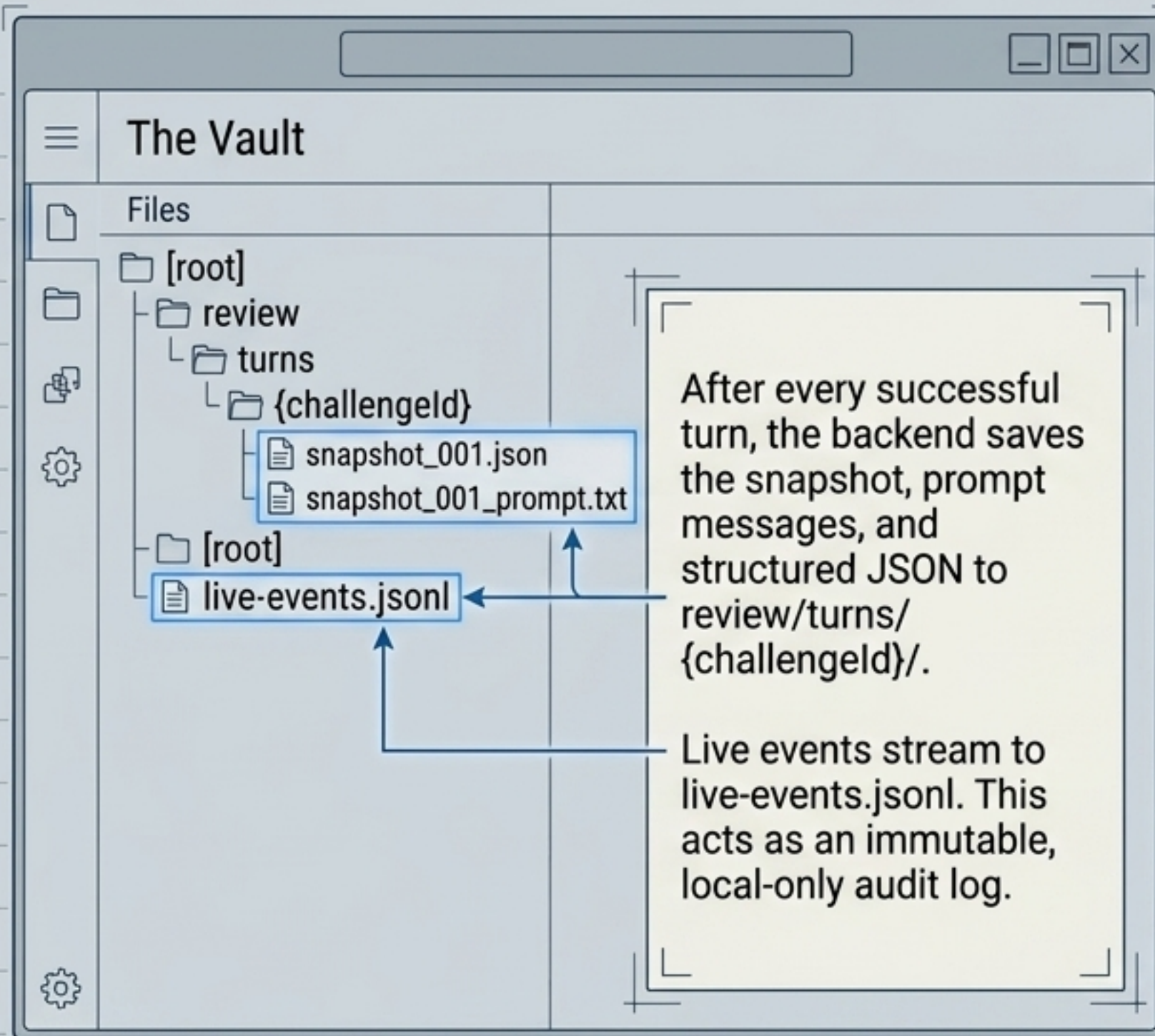
```
> ./run_arena.bat
```

(One-command launcher loading variables into both .NET and Node processes.)

Local Disk Mode: Ingestion Filter



Complete transparency for future human inspection.



Note: Both modes safely preserve the original target source folder and the immutable review logs.